# EC8D-01B

## Bistable electronic switch controller

Essential datasheet | English translation and abridgement

**2.0–5.0 V**
Operating supply voltage

**2 µA max**
Quiescent current, no load

**LOW / high-Z**
OUT states

## 1 Function and switching sequence

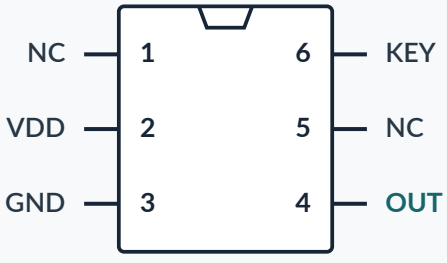
Low-power CMOS controller with an internal oscillator resistor, one pushbutton input and one active-low output. Each short button press toggles the retained output state.

| Event | OUT pin (4) | State |
|---|---|---|
| Power is first applied | High impedance | **OFF** |
| Short KEY press | LOW | **ON** |
| Next short KEY press | High impedance | **OFF** |

## 2 Electrical characteristics

| Characteristic | Min. | Typ. | Max. | Conditions |
|---|---|---|---|---|
| Operating voltage | 2.0 V | 3.0 V | 5.0 V | VDD |
| Quiescent current | — | — | 2 µA | VDD=3 V; no load |
| LED output current | — | 40 mA | — | VDS=1.0 V |
| Operating temperature | 0 °C | 25 °C | 60 °C | Ambient |

## 3 SOT-23-6 pinout (top view)




*Notch marks the top edge of the package.*

| | |
|---|---|
| **1, 5** | Not connected |
| **2** | VDD: positive supply |
| **3** | GND: ground |
| **4** | OUT: active LOW |
| **6** | KEY: button input |

## 4 Reference connections and practical meaning

KEY (pin 6): connect a momentary normally-open pushbutton to GND.

VDD (pin 2): place 100 nF (capacitor code 104) between VDD and GND.

OUT (pin 4): pulls LOW when ON; otherwise it is high impedance, not driven HIGH.

For a logic input, use a suitable pull-up; observe the receiving device's voltage limits.

Source: Shenzhen Elite Chip Microcircuit Co., Ltd., "EC8D-01B Electronic Switch IC" (2 pp.).
Translated/abridged: mechanical package dimensions and duplicated LED circuit illustration omitted.